

PHASE LOCKED LOOP

Progress Presentation

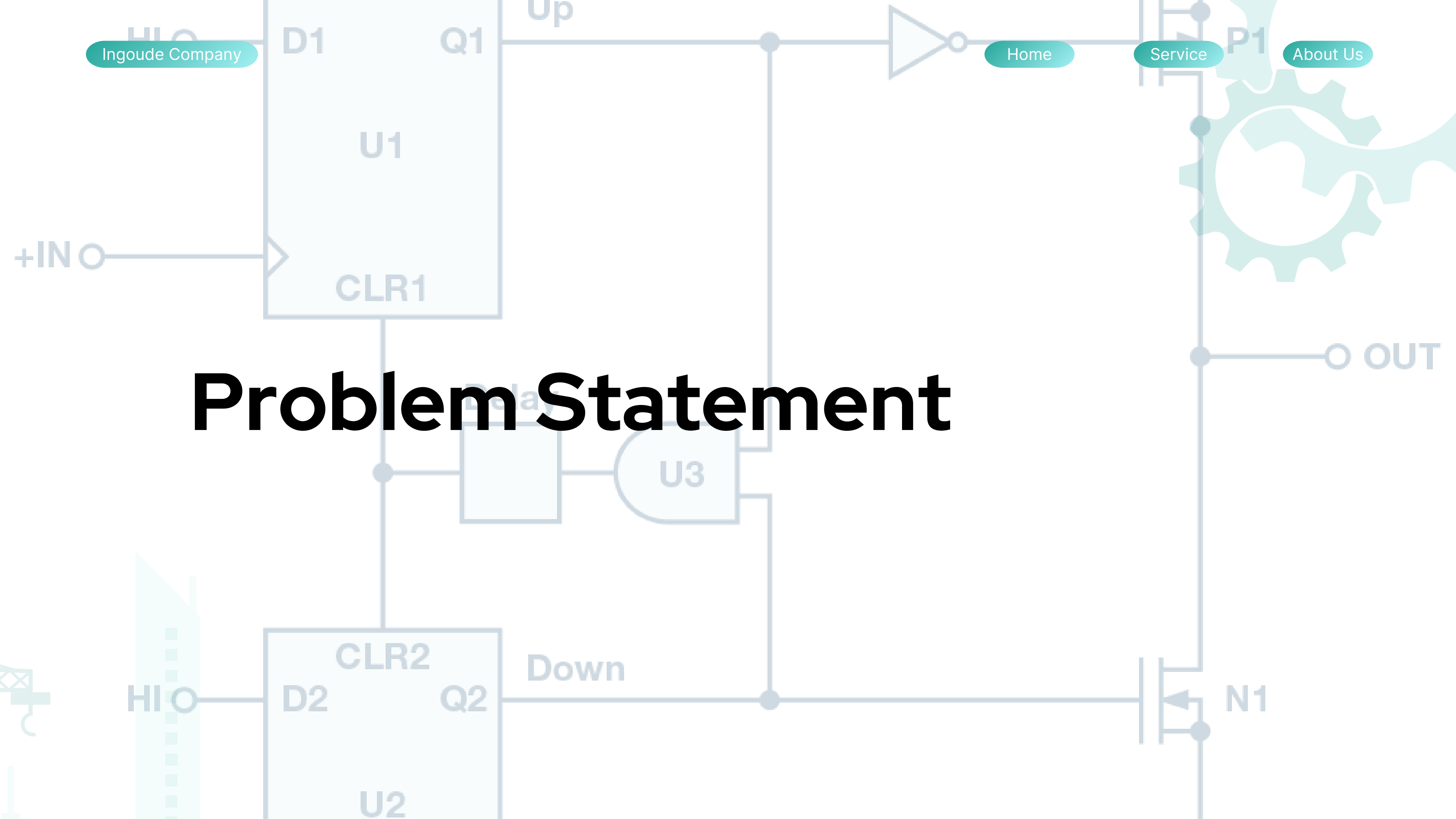
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ISHWIN KUMAR

CONTENT

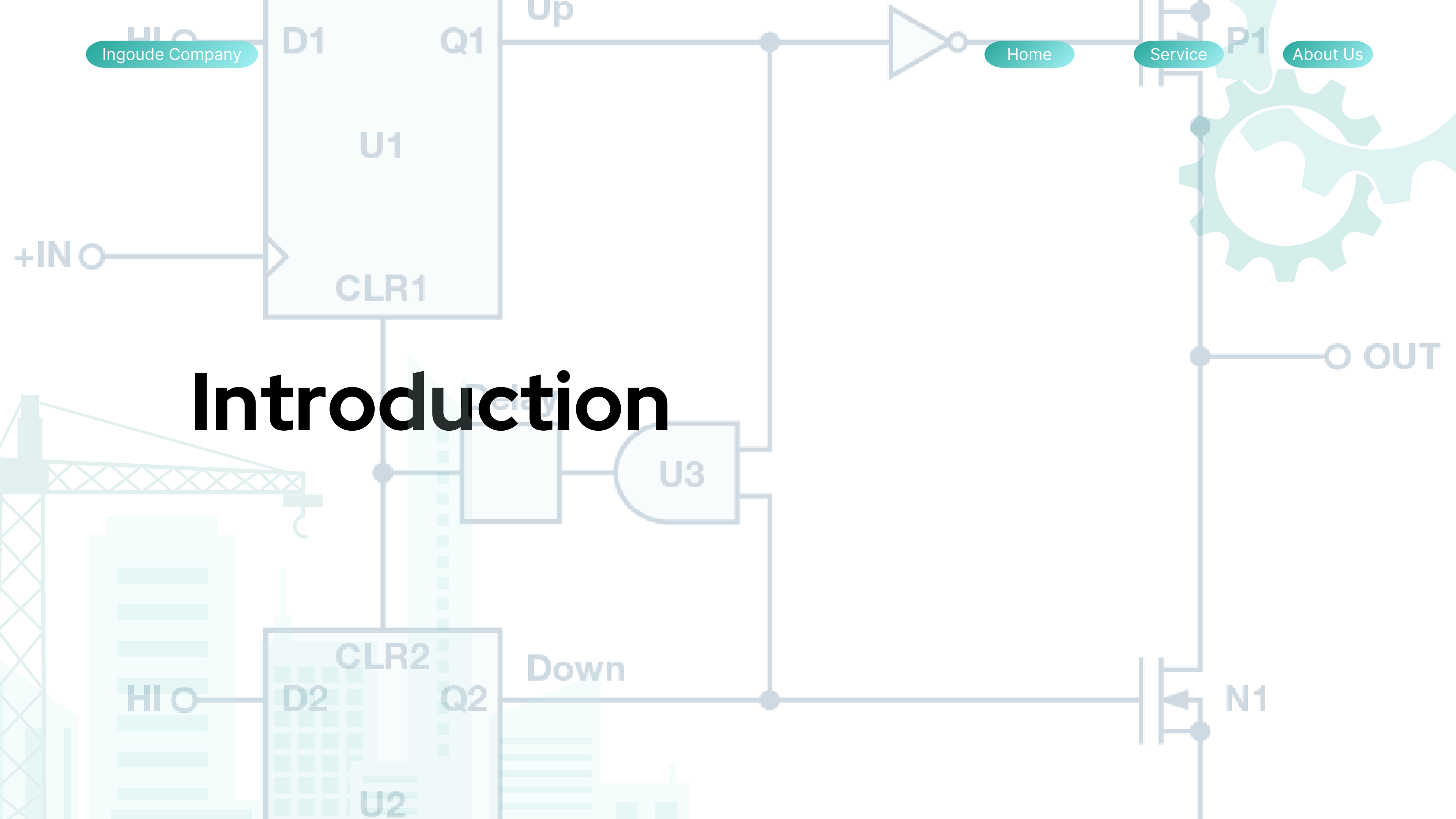
- 1) Problem Statement
- 2) Introduction to PLL
- 3) Current uses in Global scenario
- 4) Literature Review
- 5) My Progress:
 - A) *Designs made*
 - B) *Simulations*
 - C) *Hardware work till now*

Problem Statement



Design and analyze a second-order Phase-Locked Loop (PLL) by developing its key functional blocks and deriving its dynamic model. Determine critical performance parameters such as *natural frequency* (ω), *damping ratio* (ζ), *lock range*, *lock time*, and *phase margin*, and design an appropriate *loop compensator* to optimize stability of the response and overall system performance.

Introduction



What is a Phase Locked Loop?

A PLL is a negative feedback control system which generates an output signal whose phase & frequency is synchronized with a reference input. It enables precise frequency multiplication & duplication generally from a crystal oscillator in general. It continuously compares the phase difference between a reference clock and a feedback clock and adjusts the oscillator until both signals are locked.

The Building Blocks of the system

1) Phase Detector:

It is the first stage of the system and is the block that compares the phase of the Reference signal and input signal and generates an error voltage proportional to the phase difference.

2) Loop filter:

A simple RC or even an active filter which removes high-frequency components from the phase detector output and generates a DC control voltage for the VCO.

3) VCO:

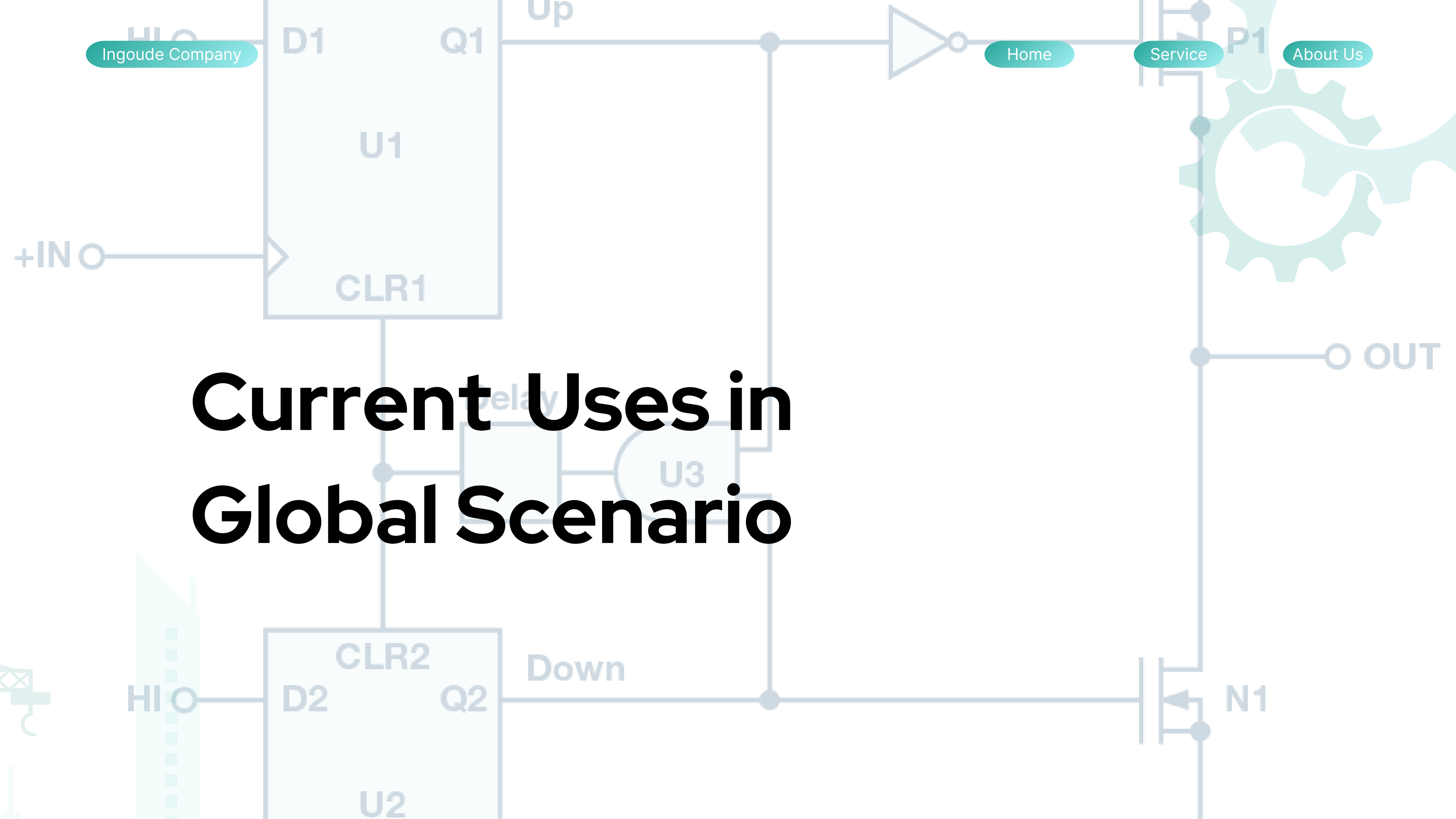
The next circuit block whose output frequency changes according to the input control voltage.

$$f_{\text{out}} = f_{\text{free}} + K_{\text{VCO}} V_{\text{control}}$$

4) FREQUENCY DIVIDER:

It is not fundamental but does the job of dividing the frequency of the VCO output by a fixed number before feeding it back to the Phase Detector.

Current Uses in Global Scenario



Power Grid Synchronization:

PLLs track the 50 Hz or 60 Hz utility grid frequency.

Grid-tied solar inverters, wind turbines, and uninterruptible power supplies (UPS) use PLLs to align their voltage phase perfectly with the grid before injecting power.

Clock Generation

Multiplies a stable low-frequency reference clock to produce high-frequency clocks required by processors, memories, communication systems, and SoCs. It provides accurate, low-jitter, and synchronized clock signals for reliable system operation.

Industrial Motor Control & Positions

Automated machinery relies on PLLs to process low-frequency encoder signals.

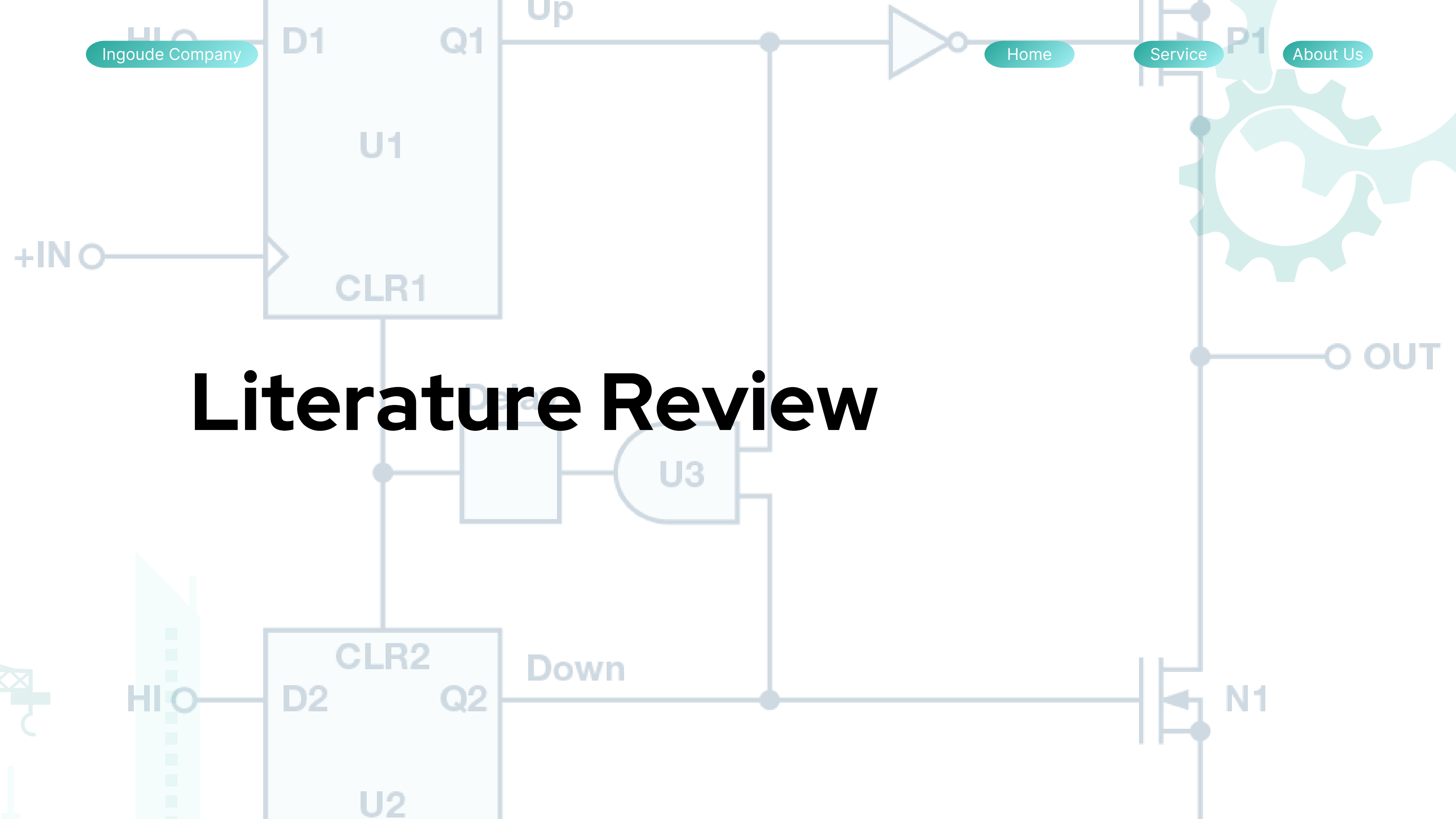
This enables the system to calculate precise speed and angular positioning for synchronous motors.

Biomedical Monitoring:

In medical sensors, PLLs monitor low-frequency biological markers such as respiration rates (0.1–0.5 Hz) and cardiac rhythm/ECG signals (1–3 Hz) to filter out background noise.



Literature Review



The linearized PLL model considered consists of a phase detector, loop filter, and voltage controlled oscillator (VCO) connected in a negative feedback loop

The forward path contains:

- Phase Detector gain: K_d (V/rad)
- Loop Filter: $F(s)$
- Voltage Controlled Oscillator gain: K_v (rad/s/V)

The loop filter is chosen as a passive lead-lag filter:

$$F(s) = R + \frac{1}{sC} \quad (1)$$

Taking a common denominator,

$$F(s) = \frac{1 + sRC}{sC} \quad (2)$$

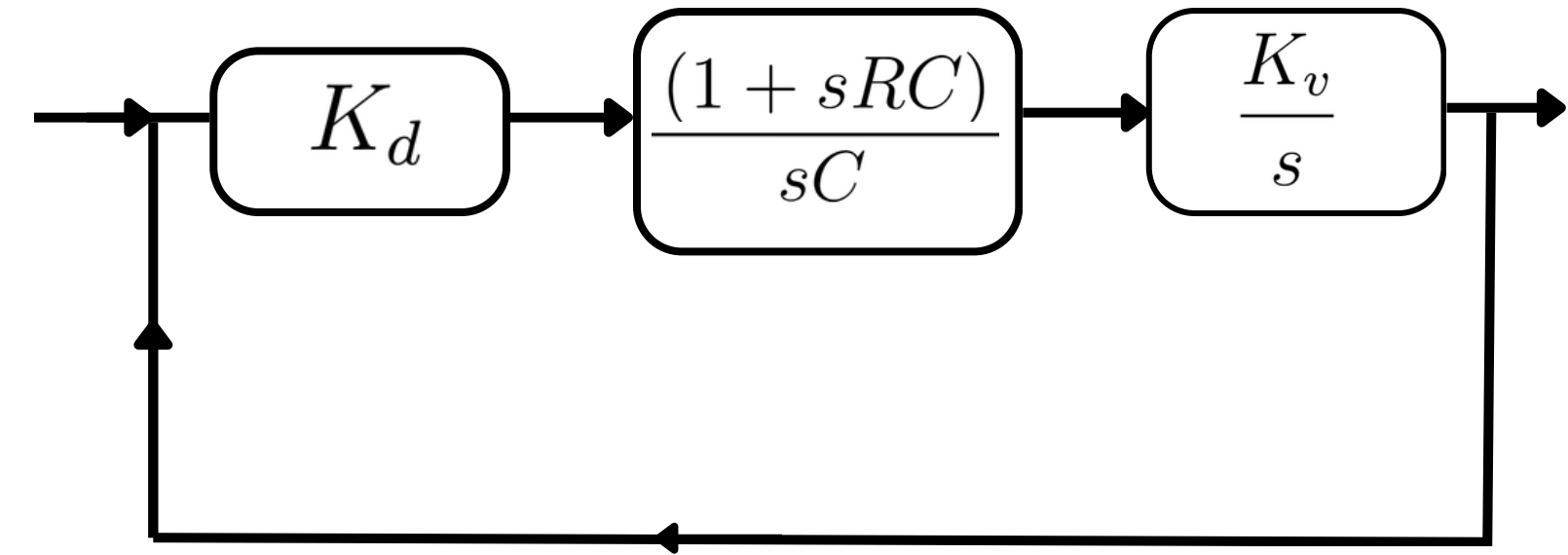
The VCO transfer function is

$$G_{VCO}(s) = \frac{K_v}{s} \quad (3)$$

since phase is the integral of frequency.

Open Loop Transfer Function

$$L(s) = \frac{K_d K_v (1 + sRC)}{s^2 C}$$



Closed Loop Transfer Function $T(s) = \frac{L(s)}{1 + L(s)}$

$$T(s) = \frac{K_d K_v (1 + sRC)}{s^2 C + K_d K_v + s R K_d K_v C}$$

The standard second-order transfer function is $T(s) = \frac{\omega_n^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

The natural frequency determines:

- Loop bandwidth
- Lock acquisition speed
- Dynamic response

Higher ω_n results in faster locking but increased sensitivity to noise.

$$\omega_n = \sqrt{\frac{K_d K_v}{C}}$$

The damping ratio determines the stability of the loop.

- $\zeta < 0.5$: Underdamped response
- $\zeta \approx 0.707$: Optimal response
- $\zeta > 1$: Overdamped response

$$\zeta = \frac{R}{2} \sqrt{K_d K_v C}$$

PLL Parameters

A) Lock Range

The lock range is the maximum frequency difference over which the PLL can directly acquire lock.

$$\Delta\omega_L \approx 2\zeta\omega_n \tag{20}$$

B) Lock-In Time

The lock-in time is the time required to achieve lock without cycle slipping.

$$T_L \approx \frac{2\pi}{\omega_n}$$

C) Pull-In Range

The pull-in range is the maximum initial frequency offset from which the PLL can eventually acquire lock.

$$\Delta\omega_P \approx 4\sqrt{\frac{2\zeta\omega_n K_d K_v}{\pi}} \tag{23}$$

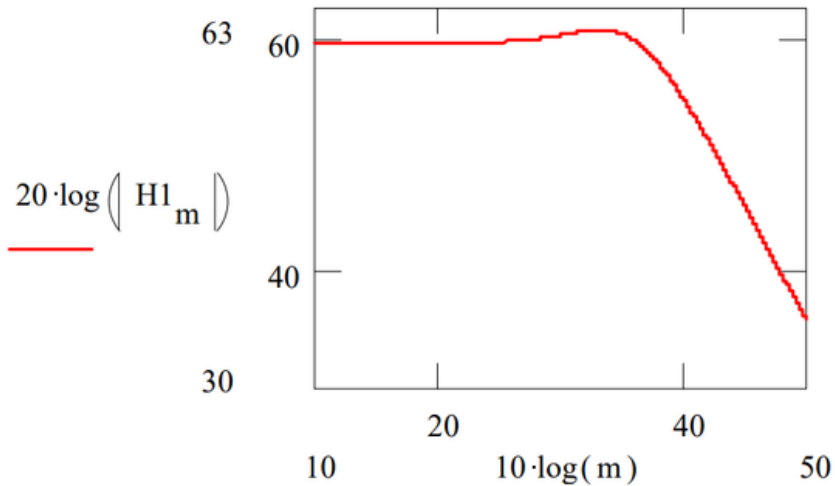
D) Pull-Out Range

The pull-out range is the maximum frequency deviation before the PLL loses lock.

$$\Delta\omega_{PO} \approx 1.8\omega_n(\zeta + 1) \tag{24}$$

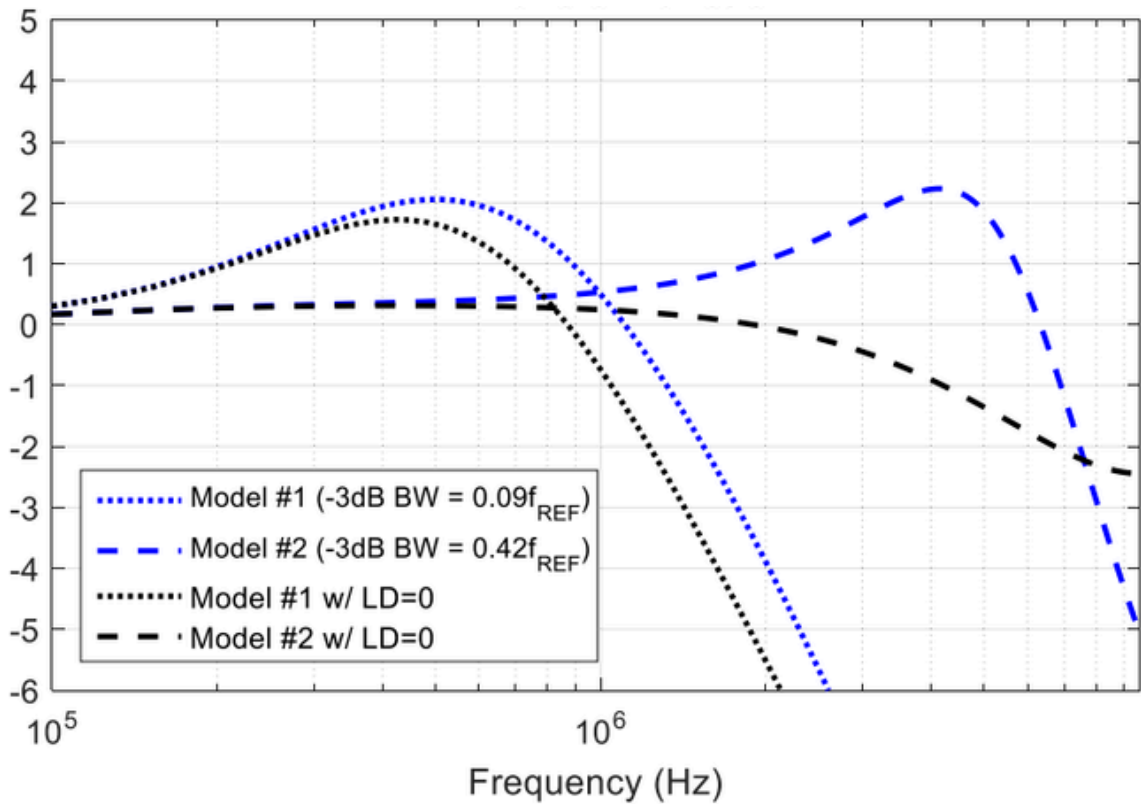
Bode Plots

Mag. Plot for Closed Loop TF for ζ=0.7



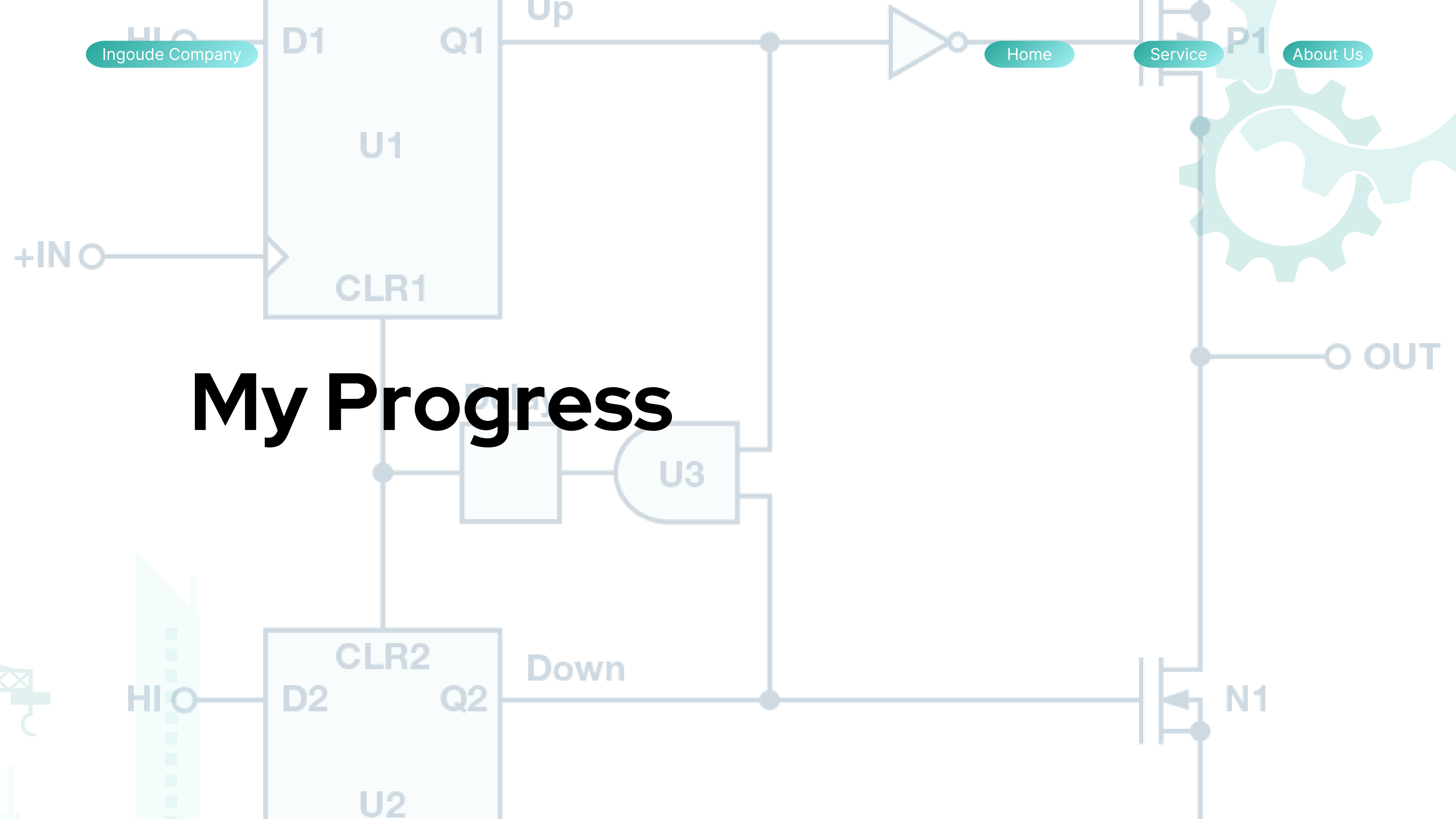
-Technical Brief SWRA029
(Texas Instruments)

CLTF for low and high bandwidth setting for 2 PLLs



-A Flexible, Low-Power
Analog PLL for SoC
and Processors in
14nm CMOS

My Progress



Designs Using MOSFETs

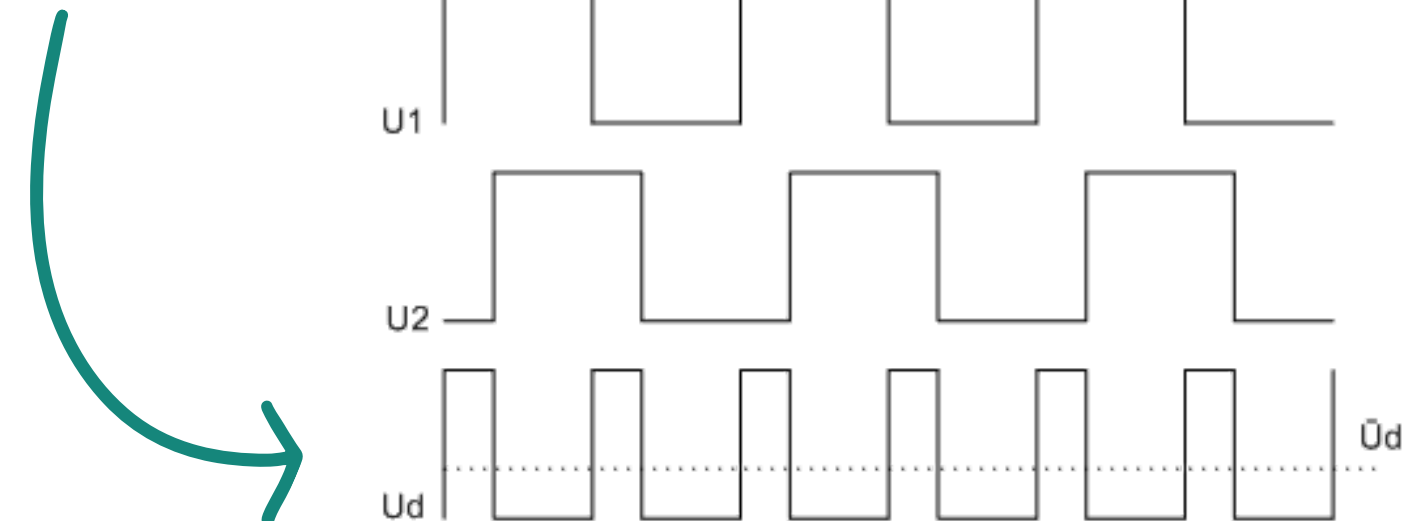
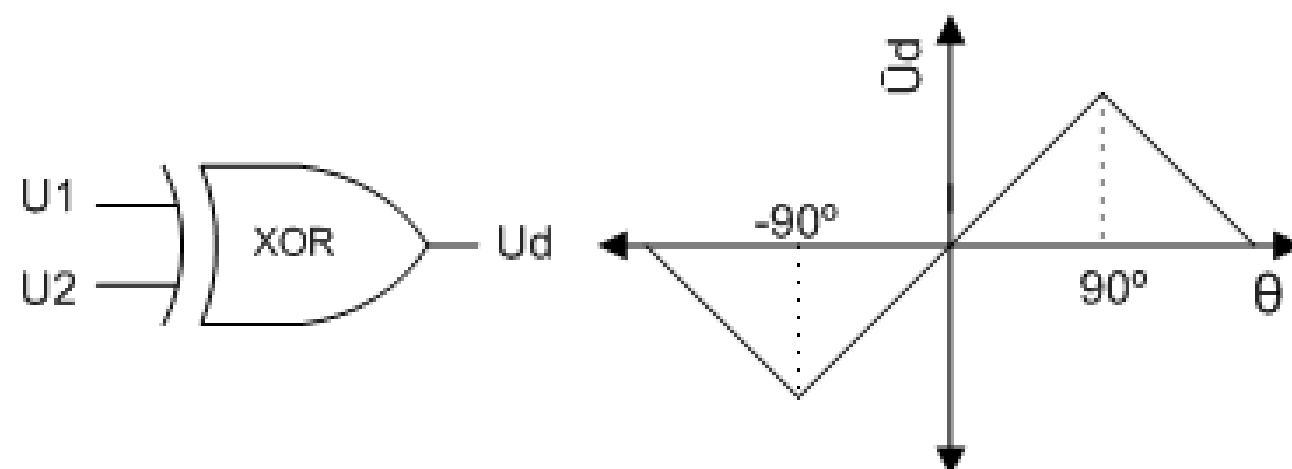
1) THE XOR GATE DESIGN

As an initial implementation, an **XOR-based phase detector** was selected which receives the reference signal and the input signal and then compares the phase between these signals using a very simple boolean logic and generates an output waveform whose **duty cycle is proportional to the phase difference** between them.

When signals are perfectly in phase, output remains low, i.e. a **duty cycle of 0%**.

When signals are exactly out of phase, output corresponds to a **duty cycle of 100%**.

As the phase difference increases, the pulse width of the output waveform increases accordingly.



XOR Gate Design (Contd..)

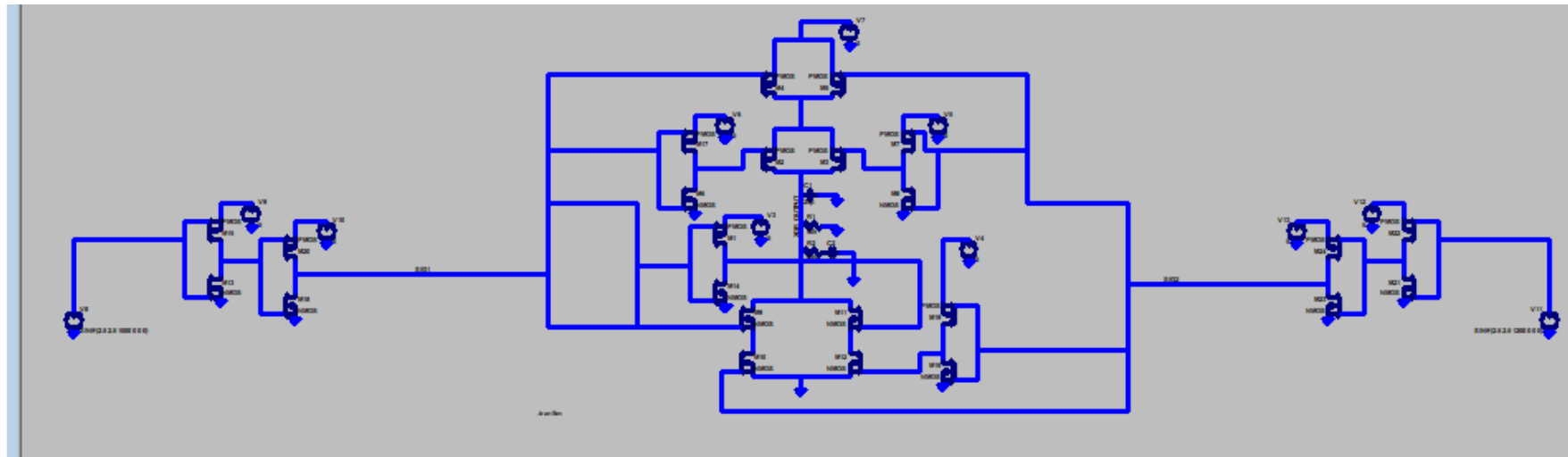


FIGURE 1

Figure 1 shows the Gate implementation using standard CMOS Inverters.

Figure 2 shows the output and input signal waveforms.

The input signals have non-zero frequency difference and offset, hence the output is expected to not be constant

The top plot is the output waveform

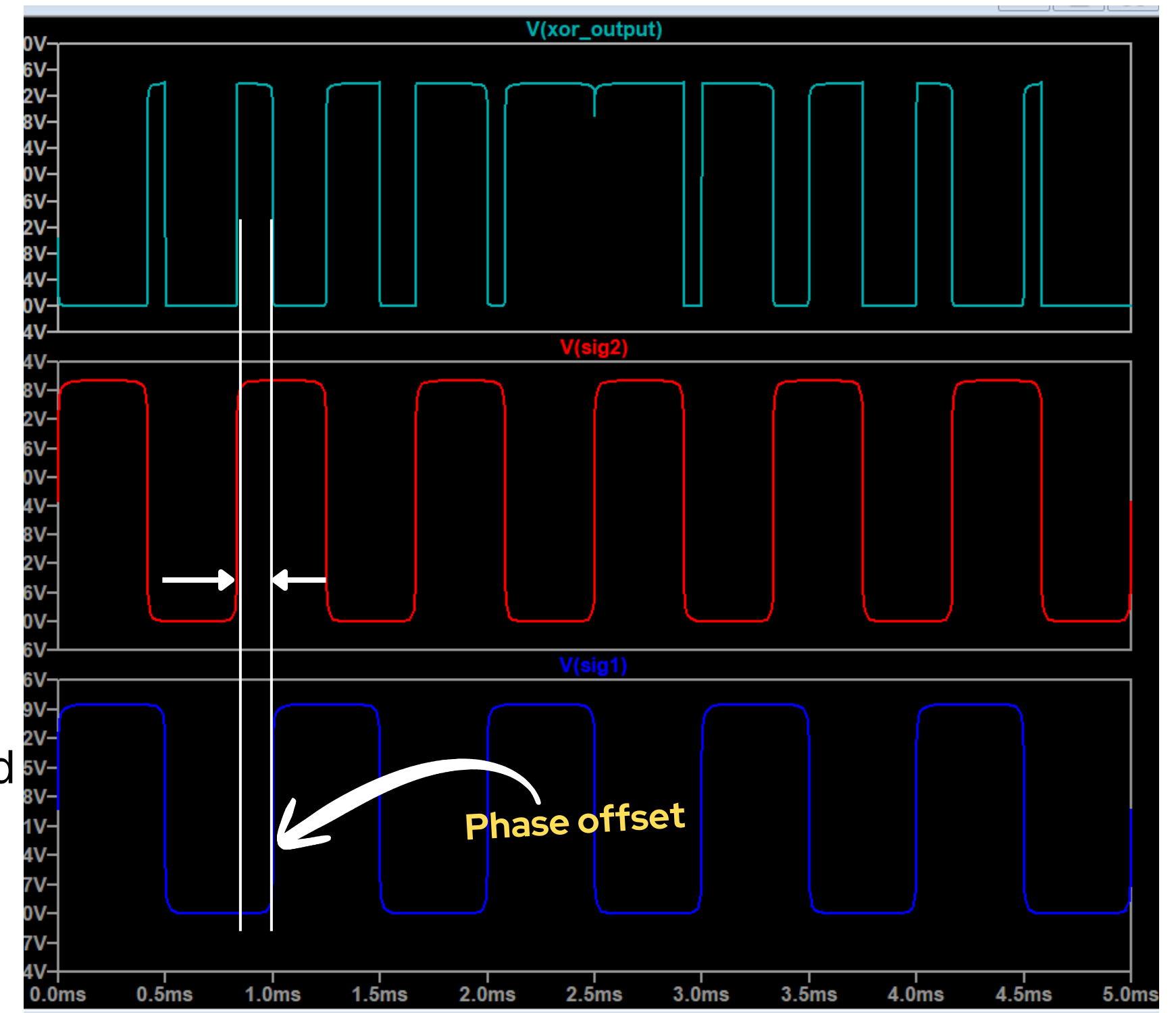


FIGURE 2

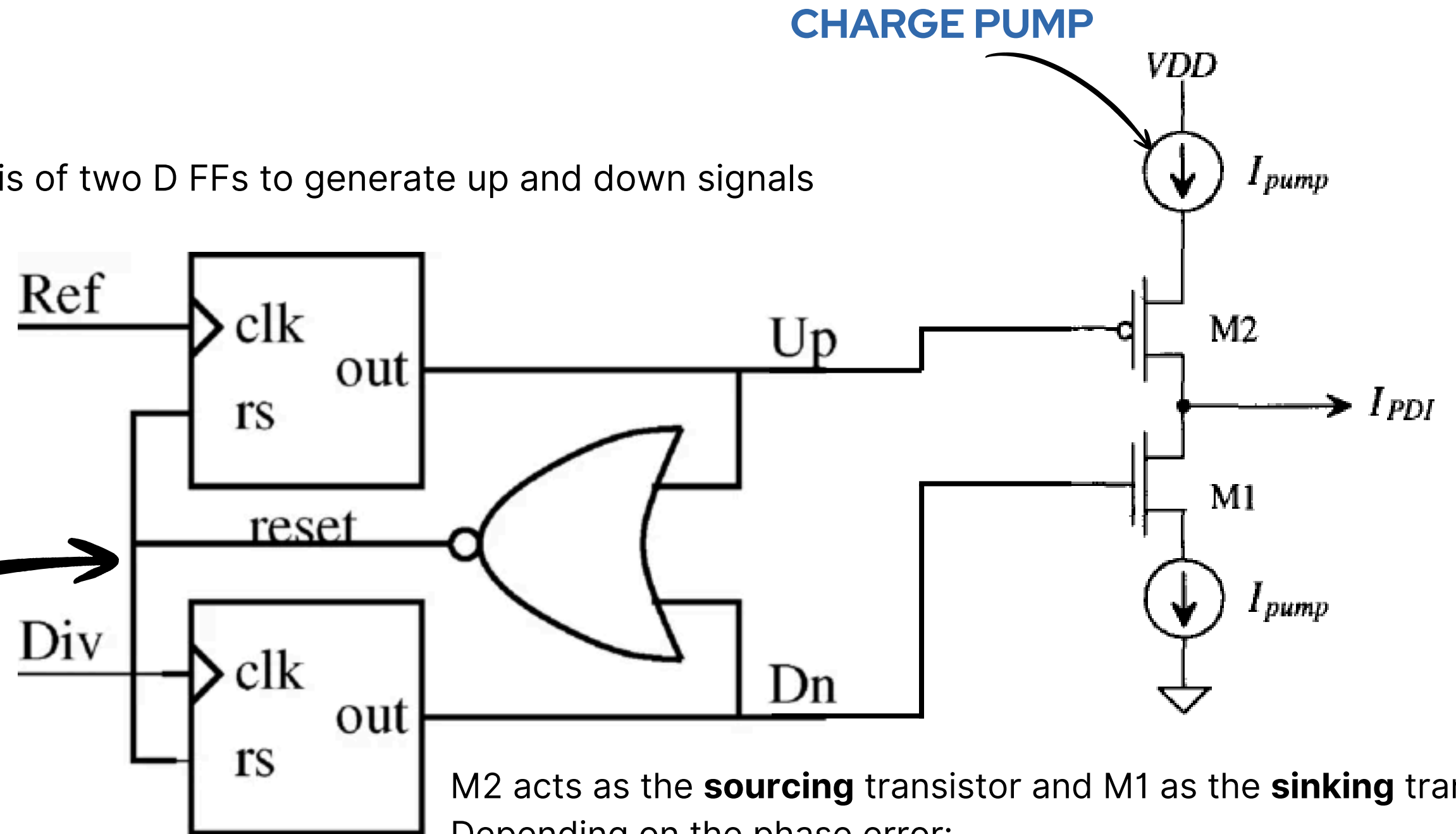
Designs Using MOSFETS

2) Using Flip Flops-

Another popular implementation usage is of two D FFs to generate up and down signals

- For lagging Input signal, **UP** signal turns **HIGH**, indicating a lagging-phase difference
- For leading Input signal, **DOWN** signal turns **HIGH**, indicating a lead-phase difference

When both signals are in phase, they are simultaneous ON, and thus, the **RESET** turns **HIGH**, making **outputs go LOW**



M2 acts as the **sourcing** transistor and M1 as the **sinking** transistor. Depending on the phase error:

UP signal active: M2 turns ON, injects current into filter, increasing the control voltage.

DOWN signal active: M1 turns ON, removes current from filter, decreasing the control voltage.

Flip Flop Design (Contd..)

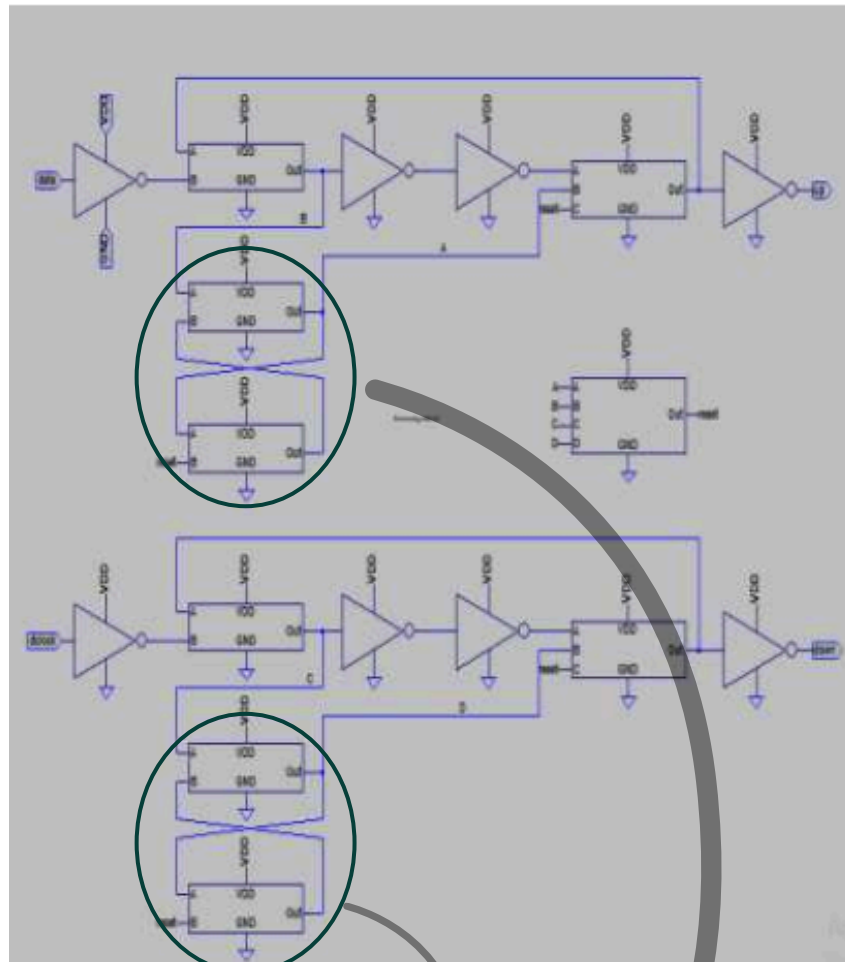


FIGURE 1: Gate level diagram

D Flip Flops

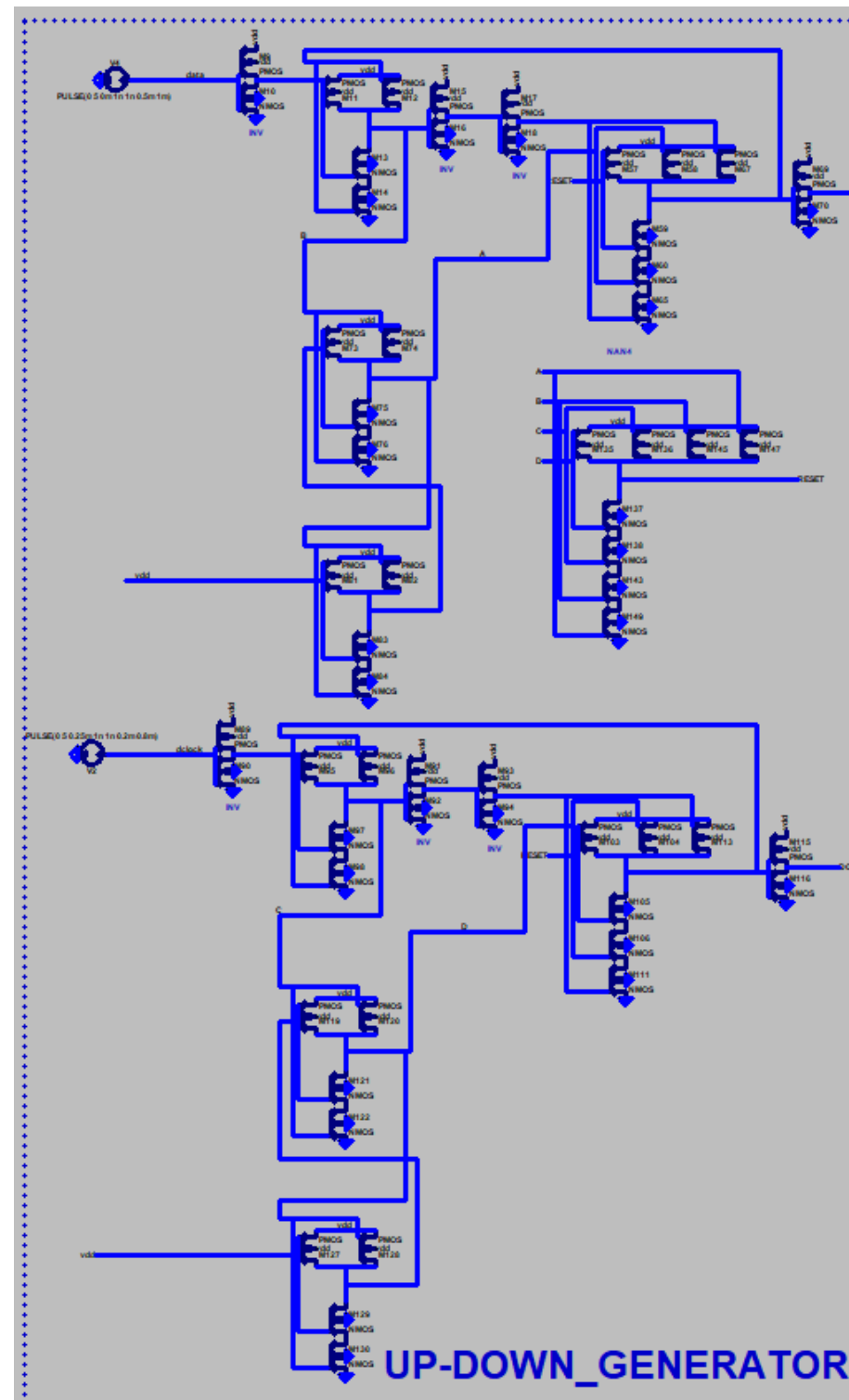


FIGURE 2: CMOS implementation using 50nm NMOS and 100nm PMOS

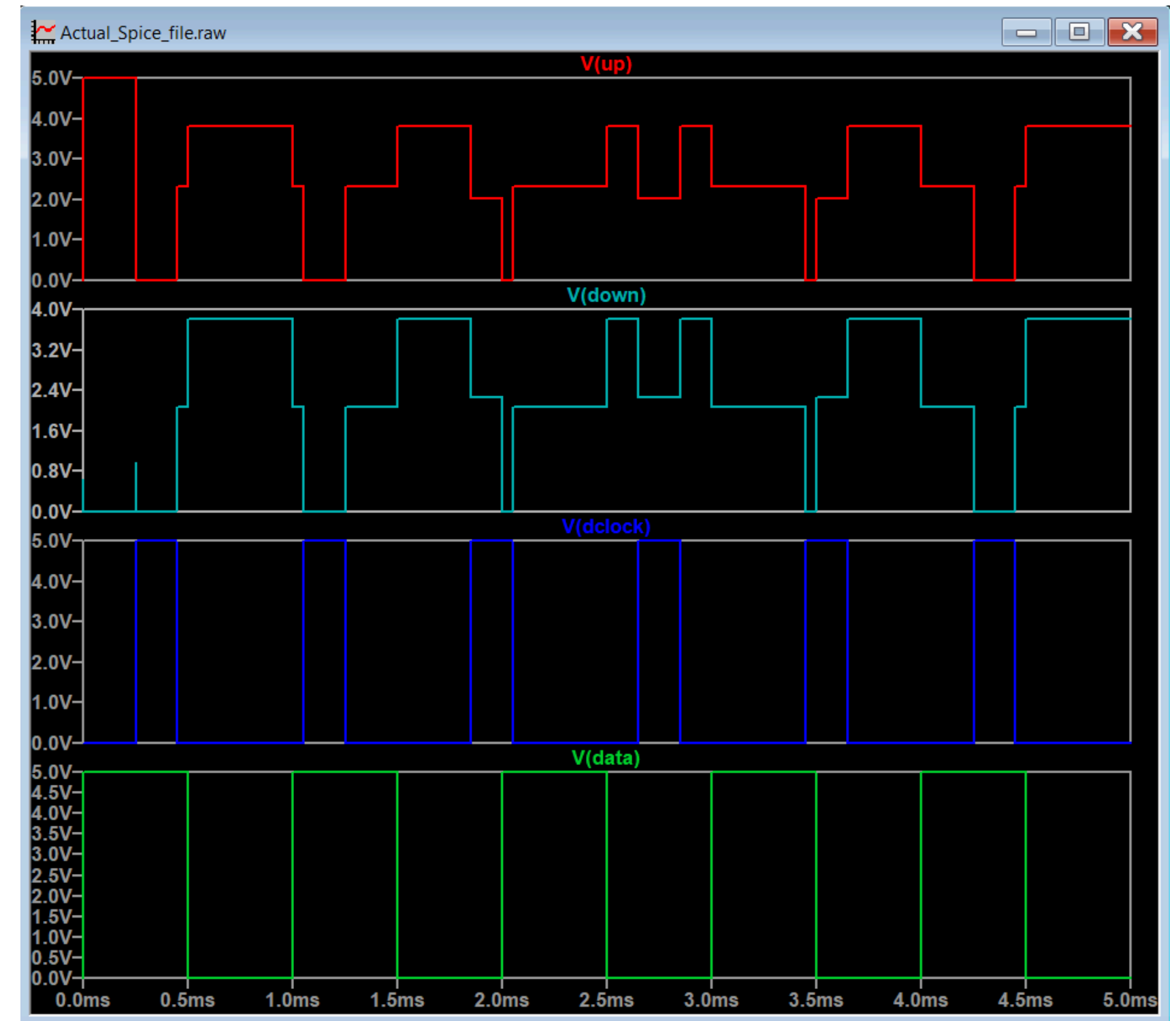


FIGURE 3: Actual Spice simulation for the generated Up & Down voltages for given signals

Flip Flop Design (Contd..)

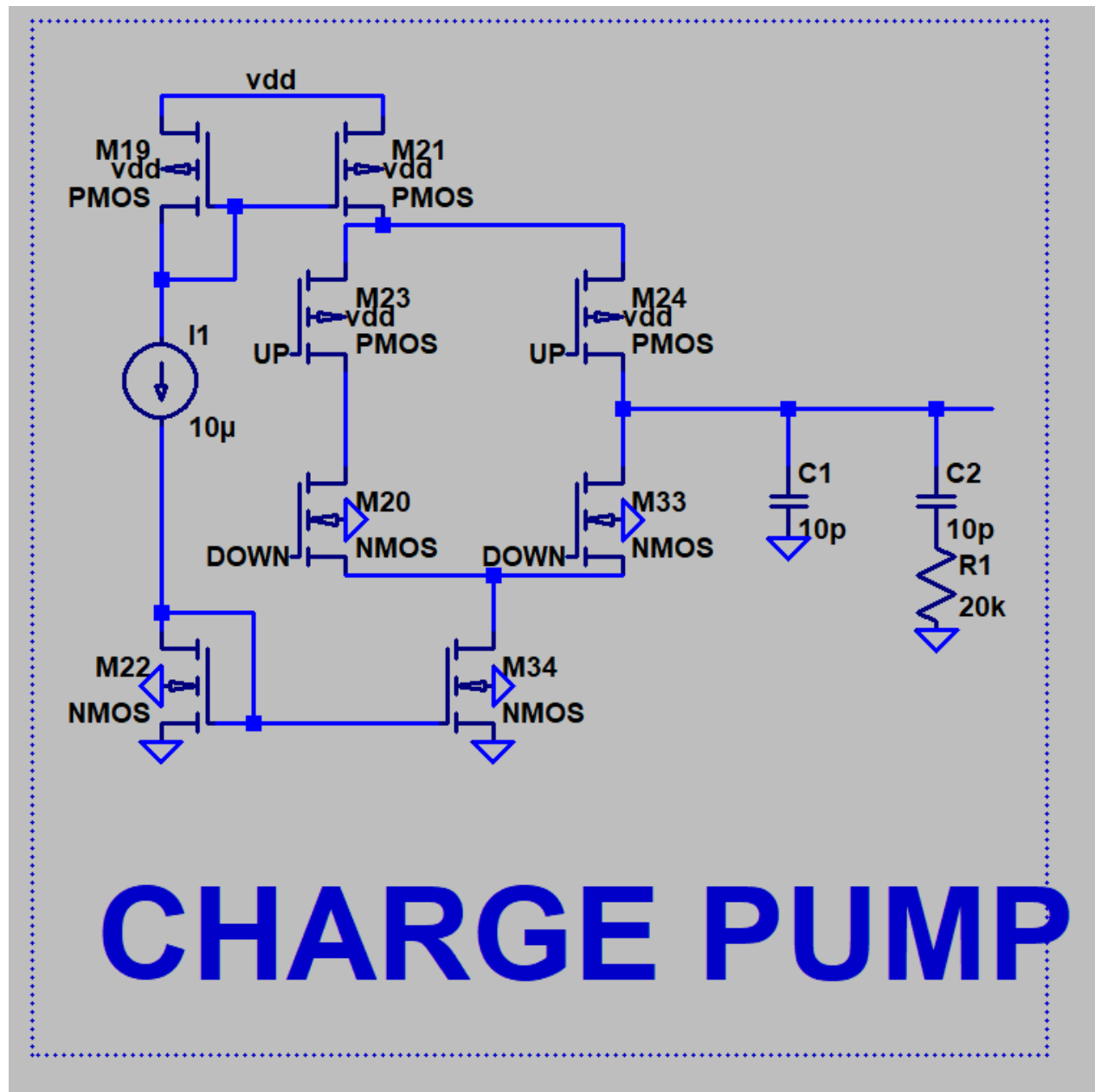


FIGURE 1: Basic charge pump with output 10p capacitor

Here, a current mirror configuration is used to bias the system amplifying the difference between UP and DOWN signals

The charge pump produces an output V_{CP} .

The generated V_{CP} pulses reflect the phase difference between the signals and serve as the control input for the PLL loop filter.

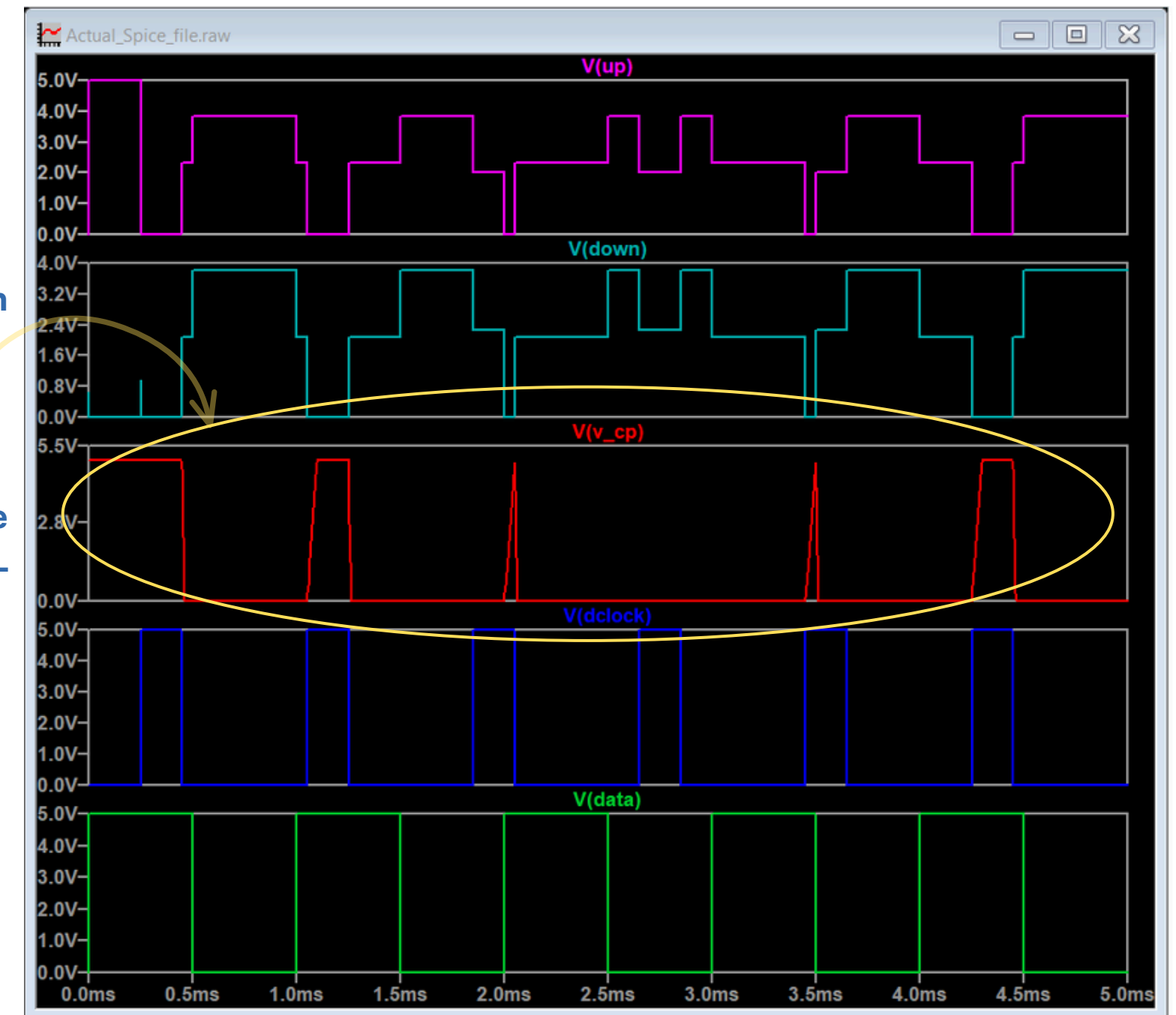
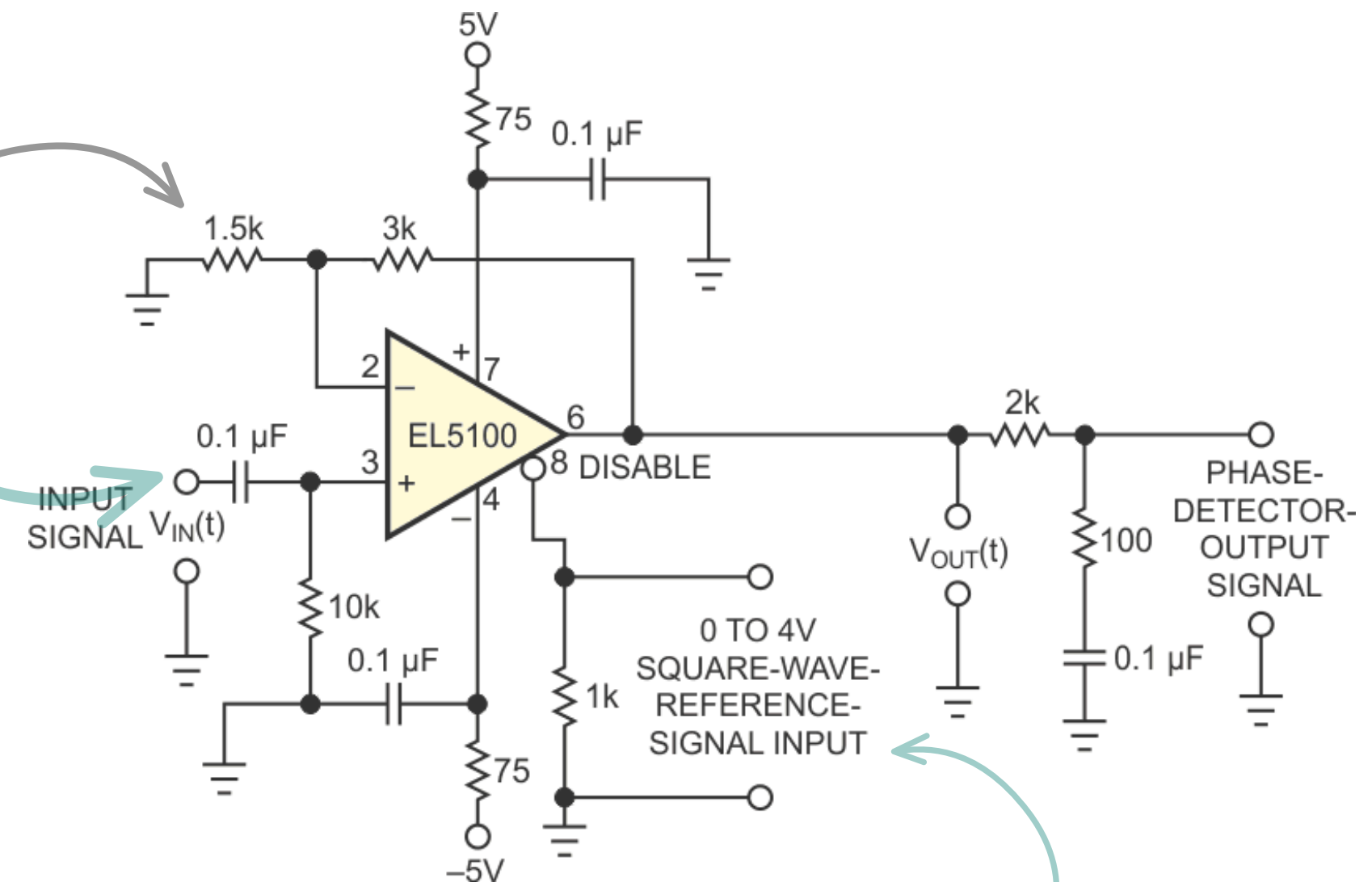


FIGURE 2: Actual Spice simulation for the final output of the phase detector (V_{CP})

Designs Using Op-Amps (Phase Detector)

THEORY

- The input signal is AC-coupled through the $0.1\ \mu\text{F}$ capacitor, **removing any DC offset**, while the $10\ \text{k}\Omega$ resistor provides **input biasing**.
- The resistor network connected to the inverting input establishes the **comparator threshold voltage**.
- When the input signal crosses this threshold, the comparator switches its output state.
- The **timing** of these switching events is then compared with the reference square wave applied to the comparator's **reference input**.
- If the input and reference signals are in phase, the output transitions occur at the same time. If one signal leads or lags the other, a phase error is created, causing changes in the width and timing of the output pulses.



The ref signal at the disable input acts as a voltage control switch, i.e. only when it is HIGH does the output passes

OpAmp based Phase Detector Design (SPICE)

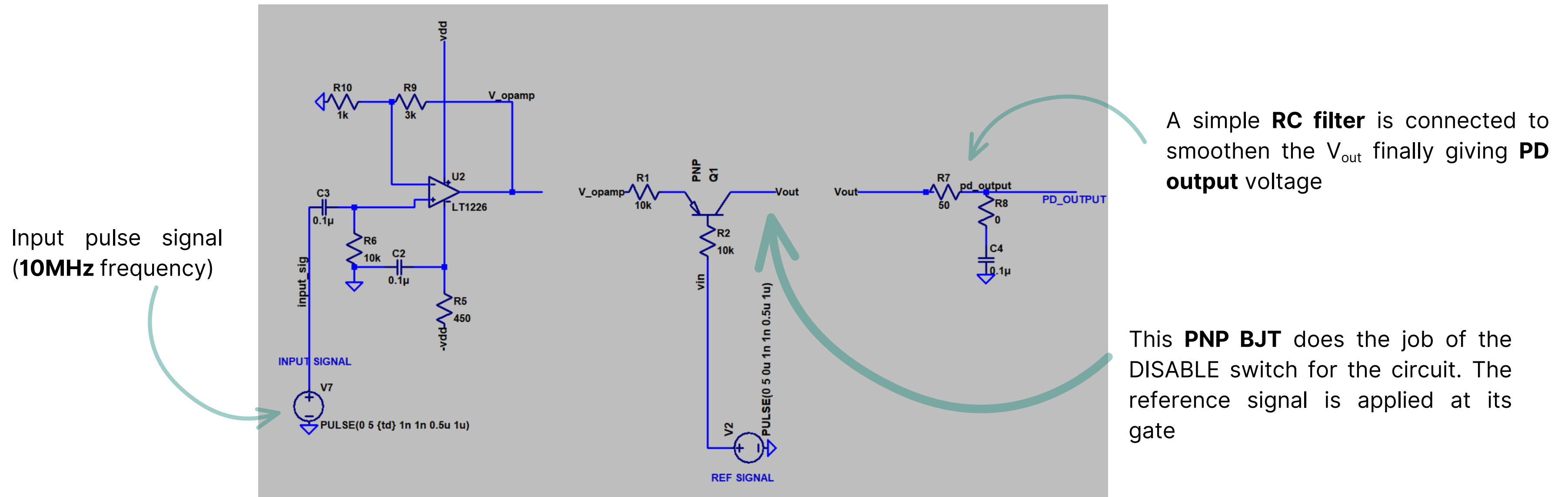


FIGURE: (From Left to right)- Main comparator using an LT1226 OpAmp, a BJT for the voltage controlled switch, RC Filter

OpAmp based Phase Detector Plots

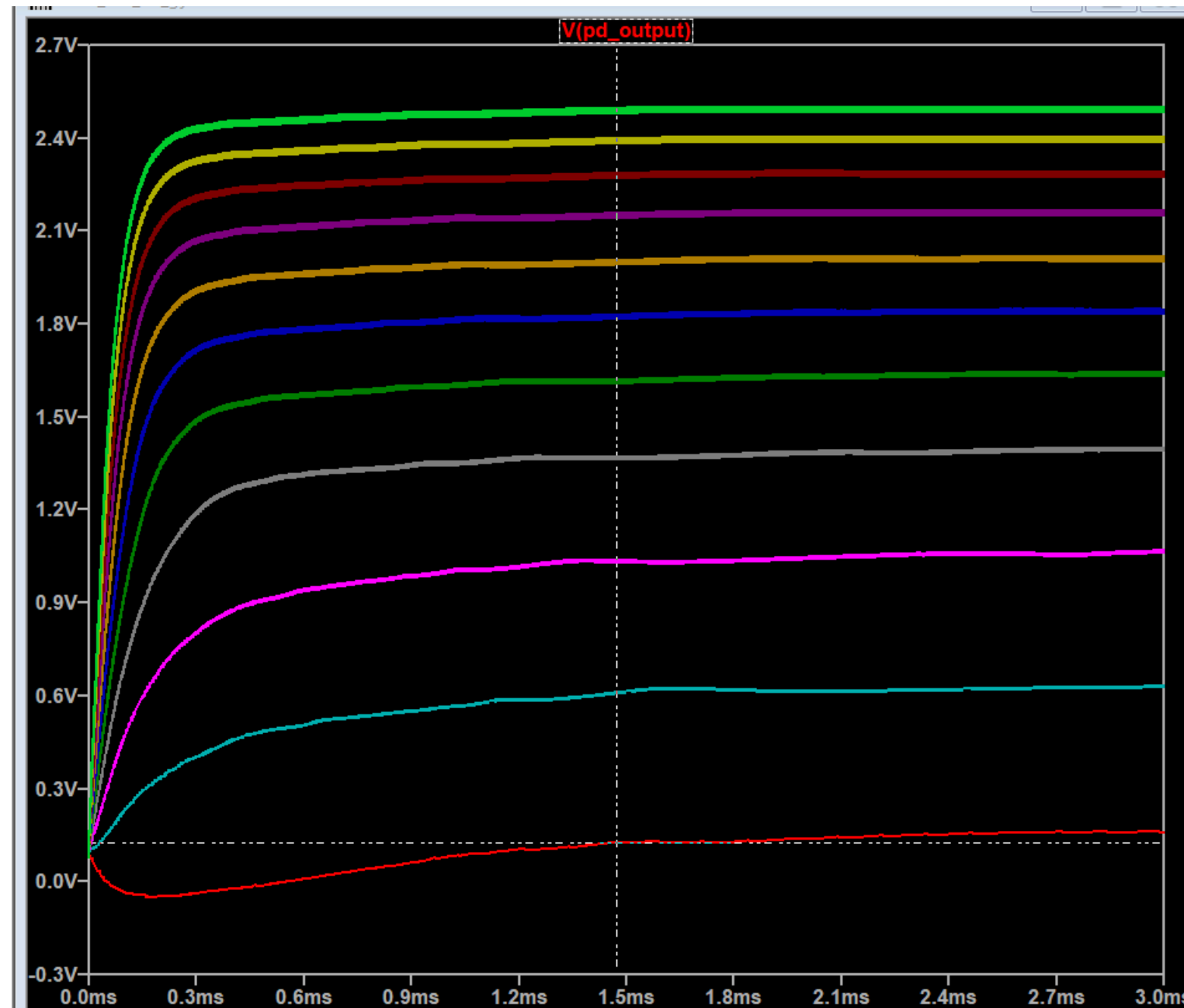


FIGURE 1

Figure 1: Plot of Phase Detector output voltage vs time for increasing Phase difference.

The phase difference varies from 0 to 360 degree.

As expected,
the voltage increases from 0 to 180 degree,
reaches a **maximum of 2.612V**

and then drops to the **initial 0.153V** at 360 degree

Figure 2 shows the plot of Output vs Phase difference. At 0.5ms time lag (i.e. 180 degree phase offset), we have the maximum output.

The best fit curve for the data points
is **$0.16 + 2.452 \sin \pi x$**

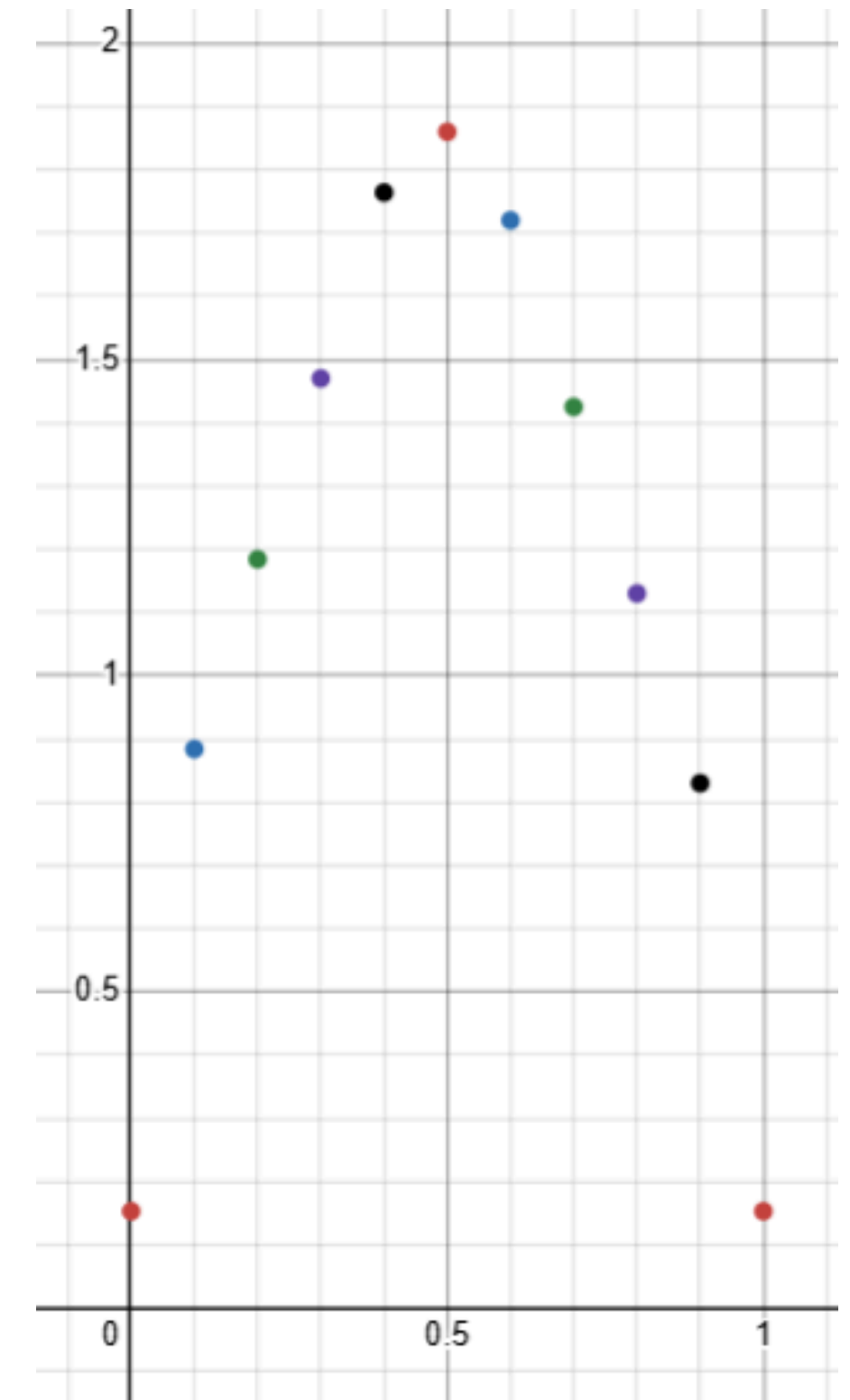


FIGURE 2

Designs Using Op-Amps (VCO)

THEORY

For Op-Amp 1,

$$V_+ = \frac{V_{in}}{2}$$

So,

$$V_- = \frac{V_{in}}{2}$$

Hence, current in the upper branch is

$$I = \frac{V_{in}}{2R_1}$$

Therefore,

$$C \left(\frac{V_{in}}{2} - V_{minus} \right) = \frac{V_{in}}{2R_1} t_{off}$$

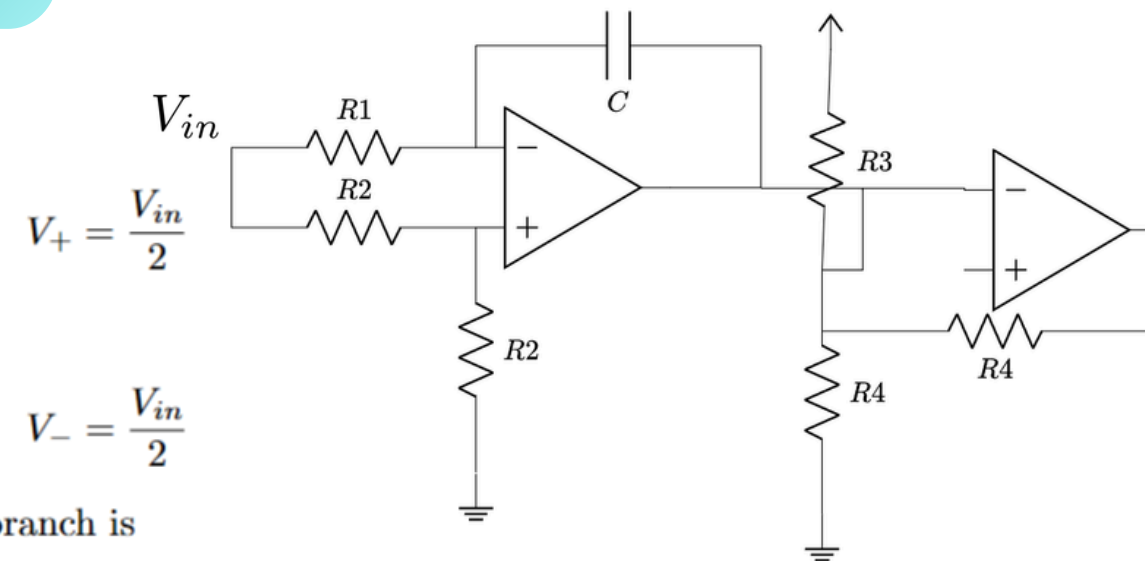
Now, t_{off} is the time until in the OpAmp2 V_{minus} reaches

$$V_{minus} = V_{DD} \left(\frac{\frac{R_4}{2}}{\frac{R_4}{2} + R_3} \right)$$

So,

$$t_{off} = 2R_1C \left(\frac{1}{2} - \frac{V_{DD} \left(\frac{R_4}{2} + R_3 \right)}{V_{in}R_3} \right)$$

FIGURE 1: Reducible circuit when $V_{minus} > V_{positive}$



(For Op-Amp 1,) $V_+ = V_- = \frac{V_{in}}{2}$

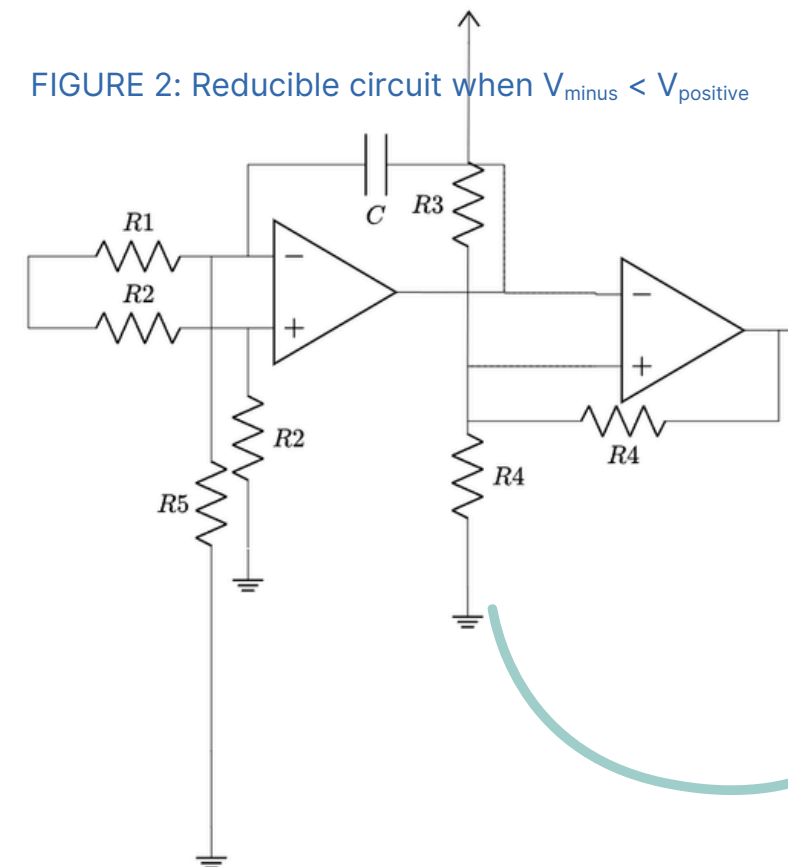
$$I = \frac{V_{in}}{2R_1} \quad (\text{through the } - \text{ve terminal})$$

$$I_{MOS} = k(V_{DD} - V_{th})^2$$

$$I_{capacitor} = k(V_{DD} - V_{th})^2 - \frac{V_{in}}{2R_1}$$

Capacitor equation: $C \left(V_- - \frac{V_{in}}{2} \right) = \left[k(V_{DD} - V_{th})^2 - \frac{V_{in}}{2R_1} \right] t_{on}$

FIGURE 2: Reducible circuit when $V_{minus} < V_{positive}$



$$t_{on} = \frac{C \left(V_- - \frac{V_{in}}{2} \right)}{k(V_{DD} - V_{th})^2 - \frac{V_{in}}{2R_1}}$$

Also, t_{on} occurs when $V_- = V_+$

$$V_+ = \frac{5}{R_4} \left(\frac{R_4}{2} \parallel R_3 \right)$$

$$t_{on} = \frac{C \left[\frac{5}{R_4} \left(\frac{R_4}{2} \parallel R_3 \right) - \frac{V_{in}}{2} \right]}{k(V_{DD} - V_{th})^2 - \frac{V_{in}}{2R_1}}$$

OpAmp based VCO Design

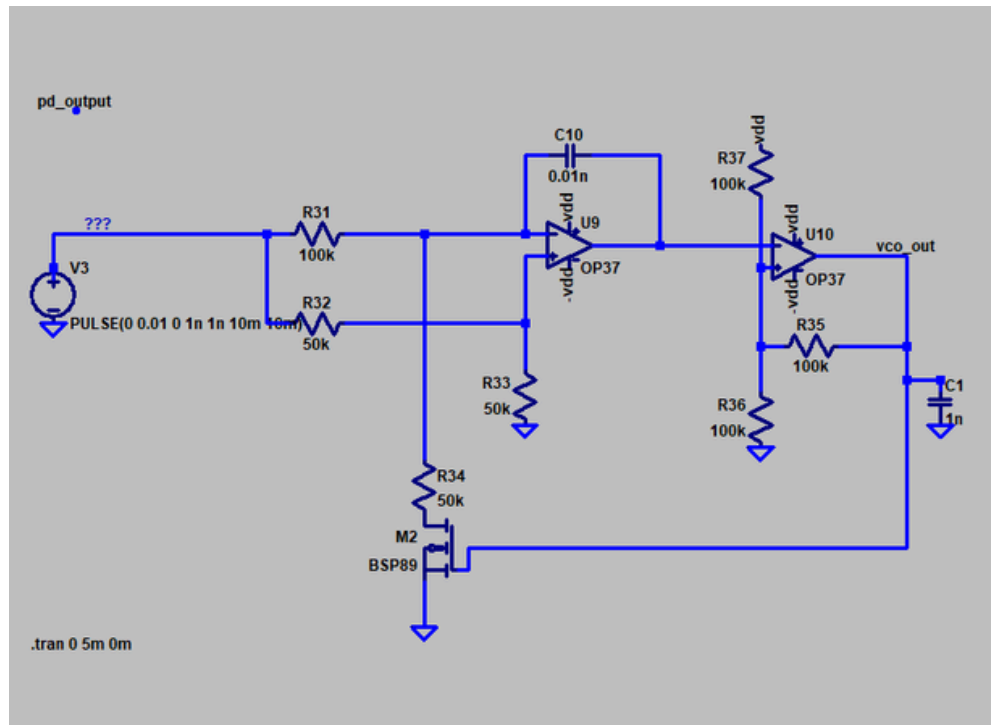


FIGURE 1: VCO circuit using OP37 OpAmp, at 100k, 50k resistors

Figure 2 Shows the linear relationship between the output signal of the VCO and input voltage.

The best fit curve for the data points is:

$$f_{out} = 1.6 + (317.46/200)V_{in}$$

Here $f_{free} = 1600\text{kHz}$

$k_{VCO} = 16.873 \text{ kHz/Volts}$

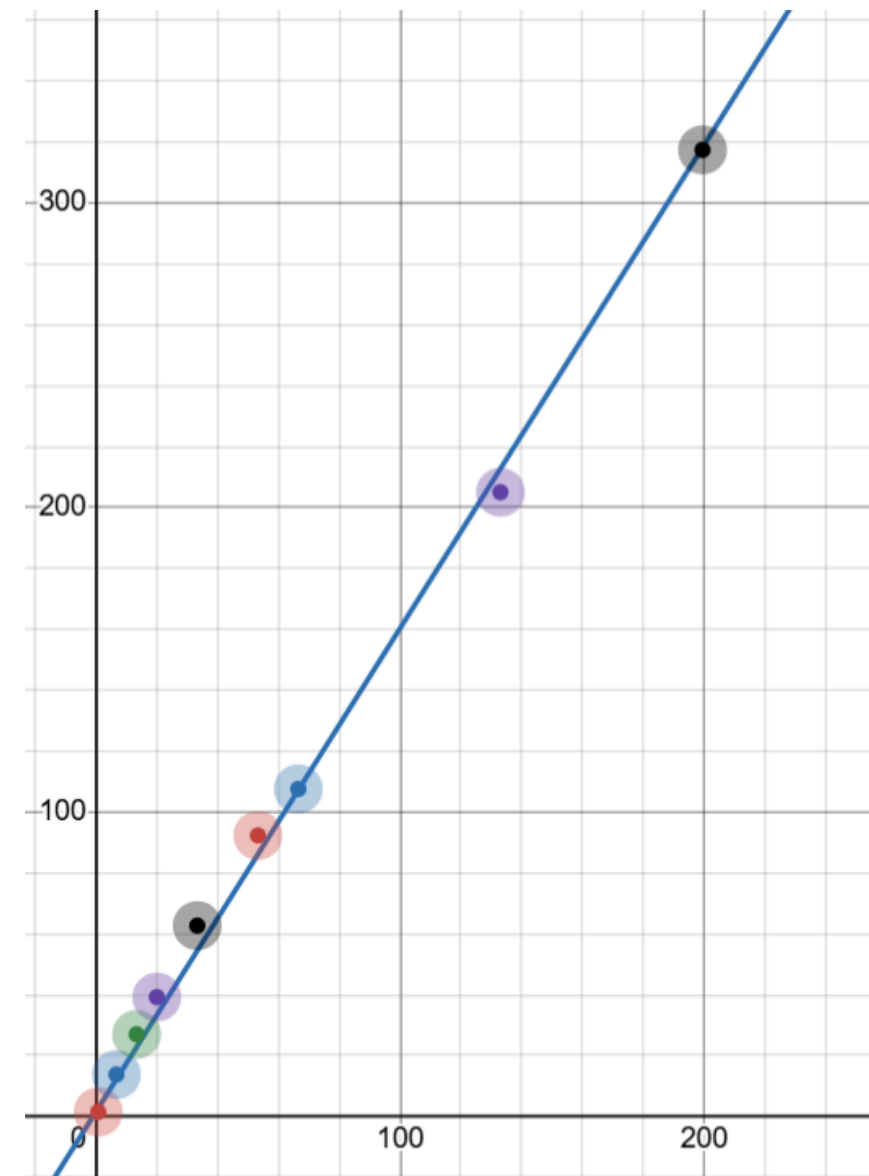


FIGURE 2: Plot of Freq_out (in kHz) vs input voltage(in 100mV)

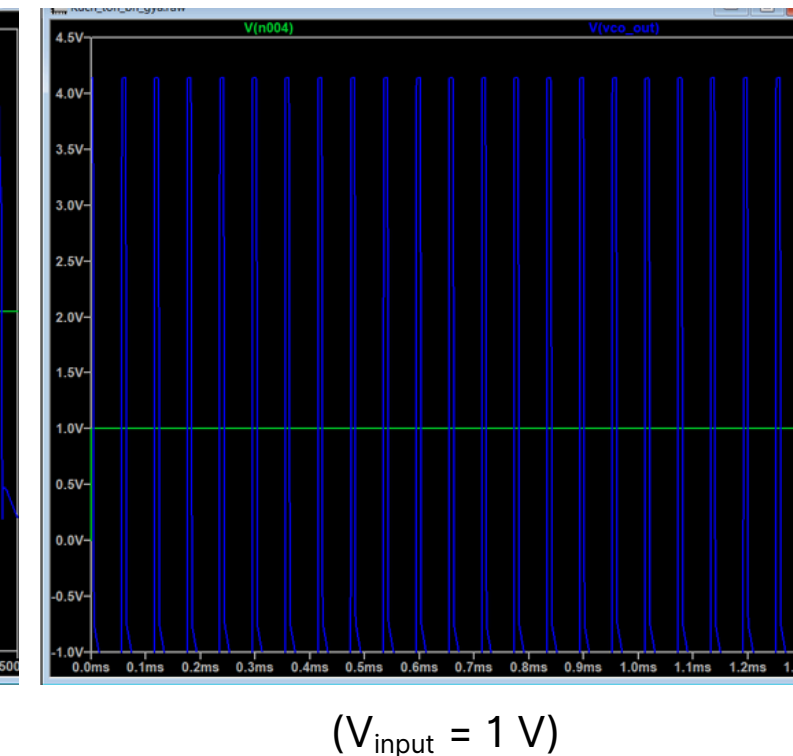
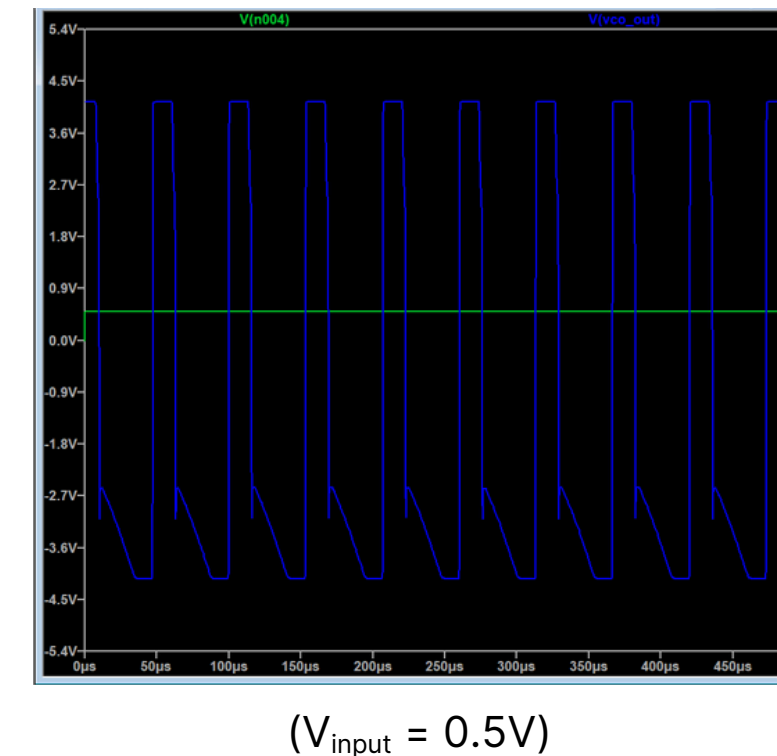
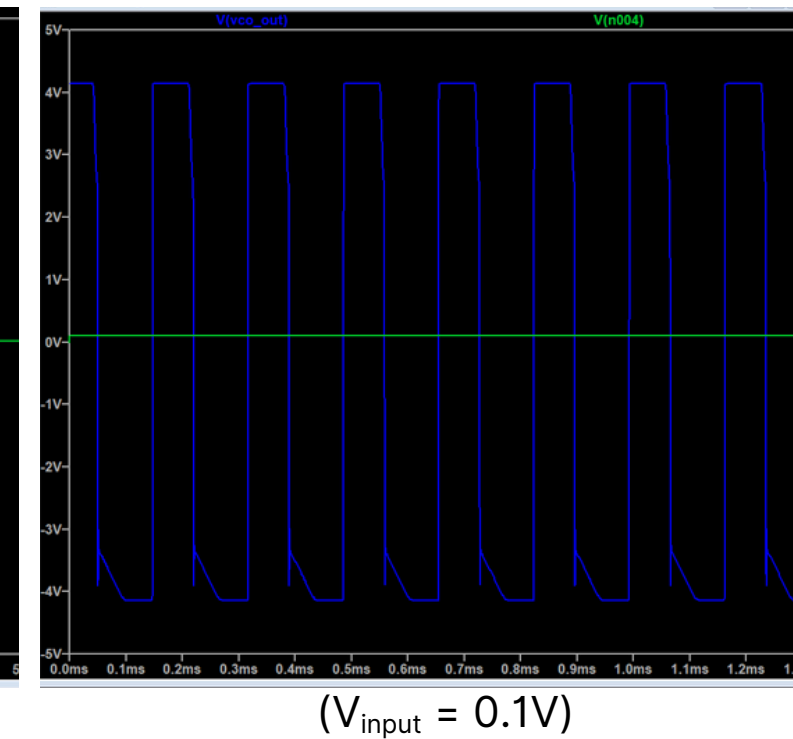
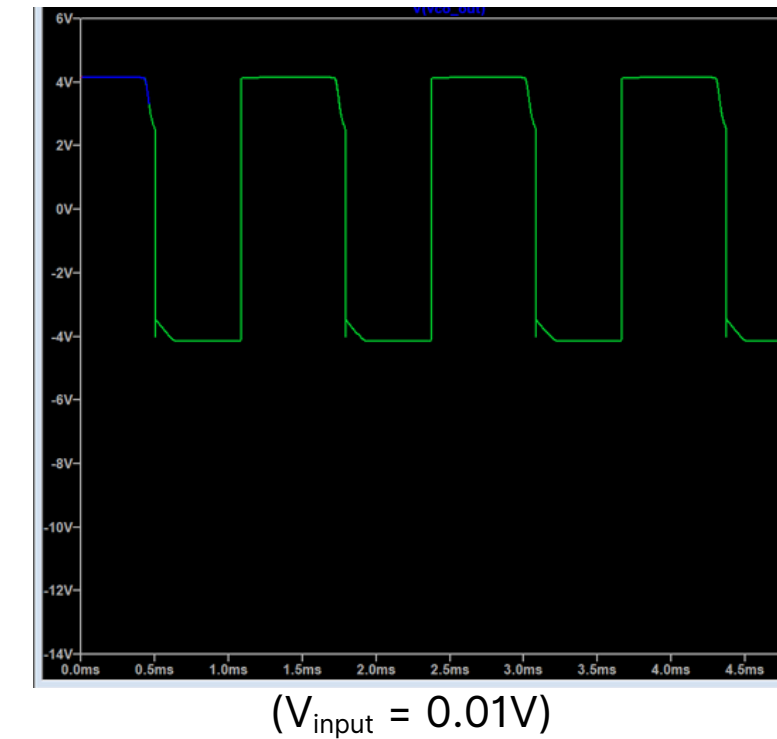
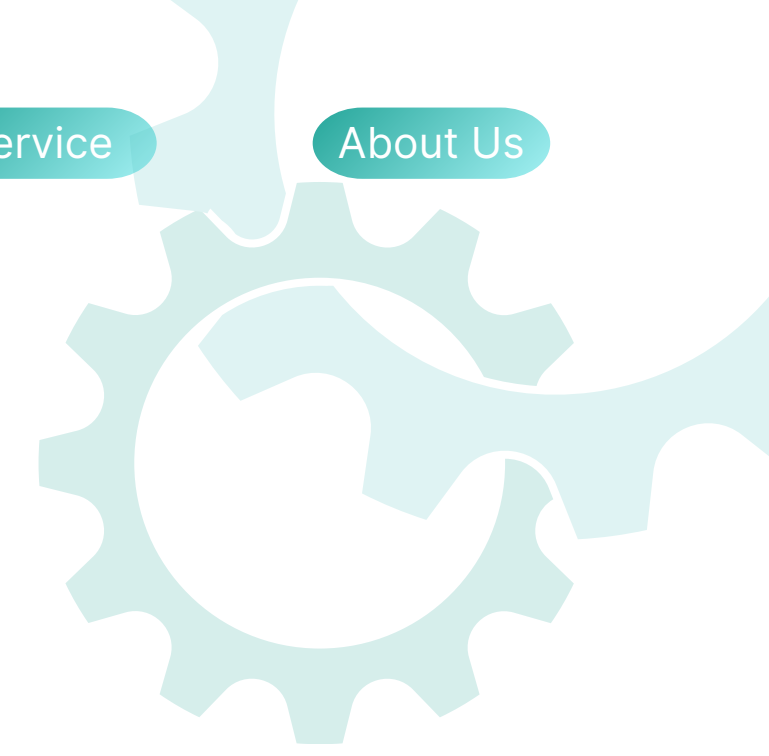


FIGURE 3: Output signals generated with increasing frequency for increasing input voltage



Thank You

